

DNA Alignment Fundamentals

BFX Workshop

09/21/2026

Sources

- Some material originally created by Jason Walker
- [Aaron Quinlan's](#) course on [Applied Computational Genomics](#):
 - Part 7: [DNA sequence mapping and alignment](#)
- The [Griffith Lab](#) course on [Precision Medicine Bioinformatics](#)
- Additional Reading:
 - [Alignment of Next-Generation Sequencing Reads](#) Knut Reinert, Ben Langmead, David Weese, Dirk J. Evers, Annual Review of Genomics and Human Genetics 2015 16:1, 133-151

The goal....

FASTQ

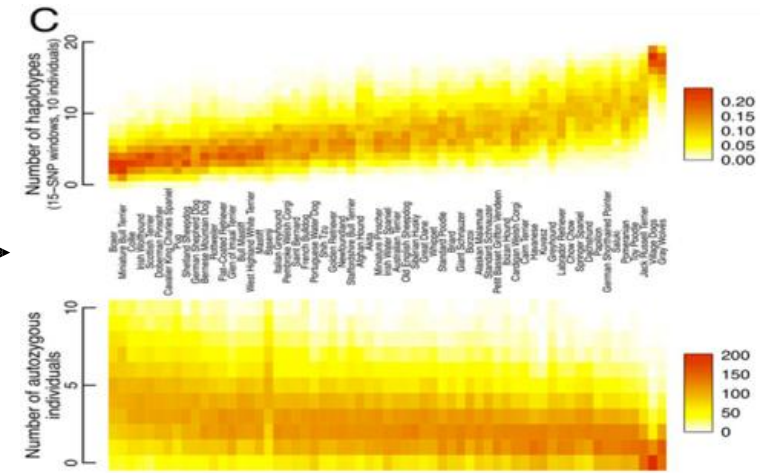
sequencing



→

cctaaccct
acccctgg
ggccctgg
actaaccct
acccctgg
cctaaccct
acccctgg
cctaaccct
acccctgg
cctaaccct
acccctgg
tcccctgg
cctaaccct
acccctgg
cctaaccct
acccctgg
cctaaccct
acccctgg

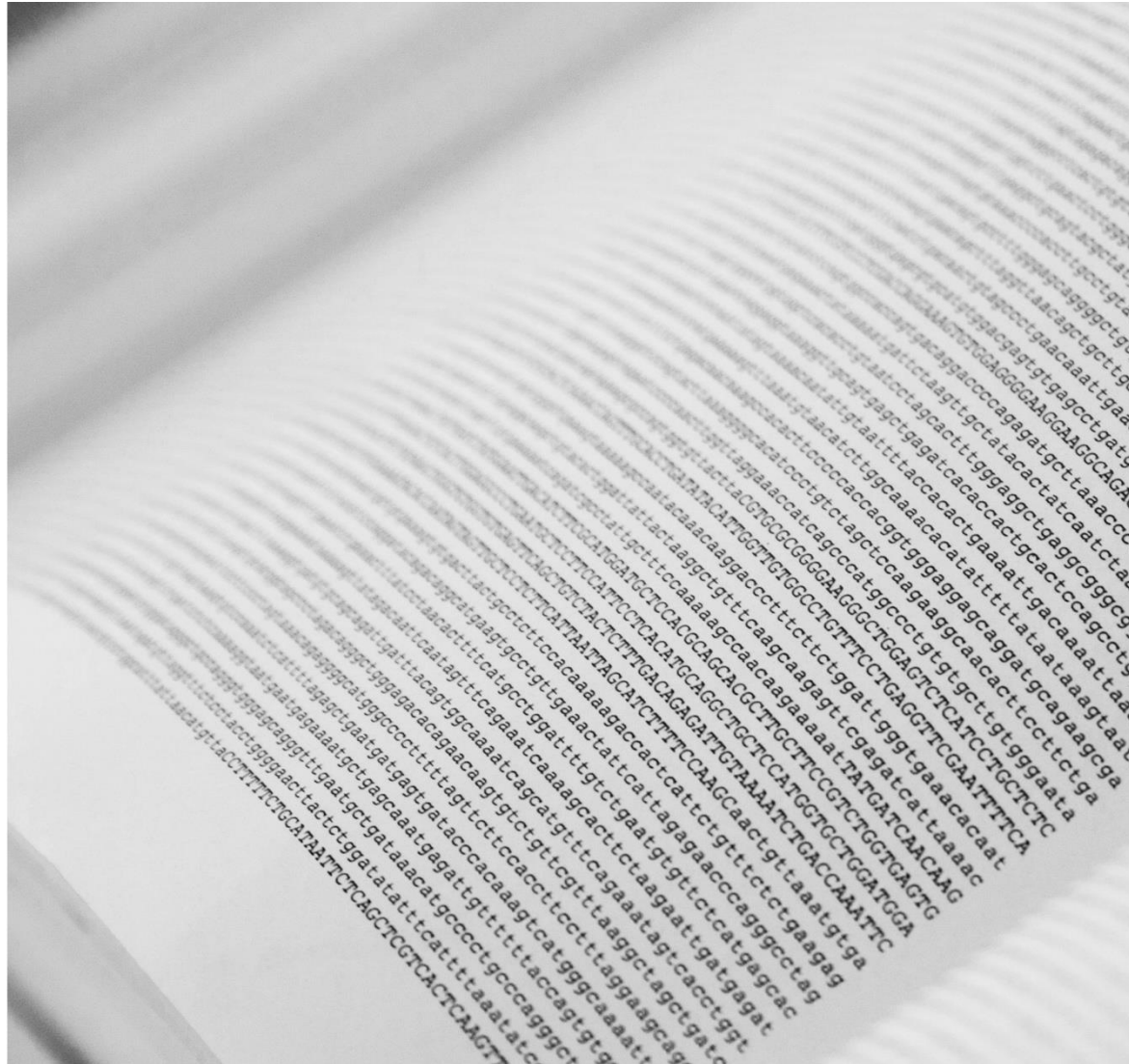
Analysis, insight, discovery



Sequence alignment is the crucial first step.

TATATATATGATAGATATGAGATATATATATATAAAAGGTACTGTAACTACTGTACATCCCGACT
TCATAATGGTGCTTTCAAACAGCGAGATGAGCAAAGACATCAGCTTCCGCCTGGCCCTCTGTG
CAAAGGGTTTCAGCCCAGG

There are ~3.2 billion bases in each copy of your DNA



Problems

A sequencer can produce over
1 billion reads per run

1 read per second
=
over 31 years

Problems

A sequencer can produce over
1 billion reads per run

About 50% of the genome is made
of repetitive sequences

```
ggcgcagagacgcaagcctacgggcgggggttgggggggctgtgtgttgca
ggagcaaagtcgcacggcgccgggctggggcggggggagggtggcgccgt
gcacgcgcagaaactcacgtcacggtggcgcggcgcagagacgggtagaat
aaccctaaccctaaccctaaccctaaccctaaccctaacccta
accctaaccctaaccctaaccctaaccctaaccctaaccctaacc
cctaaccctaaccctaaccctaaccctaaccctaaccctaaccct
taaccctaaccctaaccctaaccctaaccctaaccctaaccctaa
ccccctaaccctaaccctaaccctaaccctaaccctaaccctaacc
ccctaaccctaaccctaaccctaaccctaaccctaaccctaaccct
cccaaccctaaccctaaccctaaccctaaccctaaccctaaccct
ctaccctaaccctaaccctaaccctaaccctaaccctaaccctaacc
taaccctaaccctaaccctaaccctaaccctaaccctaaccctaacc
aaccctaaccctaaccctcgcggtaccctcagccggcccgccgcccggg
tctgacctgaggagaactgtgctccgccttcagagtaccaccgaaatctg
tgcagaggacaacgcagctccgccctcgcggtgctctccgggtctgtgct
gaggagaacgcaactccgccggcgagggcgagagaggcgccgcccgcg
gcgcaggcgagacacatgctagcgcgctcggggtggaggcgtggcgagg
cgagagaggcgccgcccggcgagggcgagagacacatgctaccgc
gtccaggggtggaggcgtggcgagggcgagagaggcgaccgcccggc
gcaggcgagagacacatgctagcgcgctccaggggtggaggcgtggcgca
```

(first bit of human chromosome 1)

Problems

A sequencer can produce over
1 billion reads per run

About 50% of the genome is made
of repetitive sequences

Sequencers make lots of mistakes

Error rate of ~0.35%

1.1 million errors in
single genome pass

Best case scenario: an error-free sequencing technology

ATTCGAAACA
TTCGCGCAAT
CTGGACTCAA



ATTCGAAACA
TTCGCGCAAT
CTGGACTCAA

Aligner

TACCTCCAGGGGGGCATCCTCCCCCCCA**ATT****CG**
AAACACAATCGTAGCCCCTGGCACTACCTATG
TGTGTCAATTCGGAGAGAGAGAGATTACGAA
AAAAAAGT**CTGGACTCAA**CTAGGATACACACA
TTCGGCTACAGATACCAAAAAAAAAAAAAAAAAA
AAATTTTCACCATTGAGGCACCACCTTCTCGT
CGCTGCGTCGCTCTGCTCGCTTCGGCTAAAAA
TTCGCGCAATACATTTCGGCTACAGATACCAAA
AAAA

Computers are rather good at finding ***exact*** matches

Reality check. Errors happen. Frequently.

ATTCGAAACA



ATT**T**GAAACA

Aligner

TACCTCCAGGGGGGCATCCTCCCCCCCA**ATT**C
AAACACAATCGTAGCCCCTGGCACTACCTATG
TGTGTCAATTCGGAGAGAGAGAG**ATT**G**GAAAC**
AAAAAAGTGCTACAGATAACCACTAGGATACAC
ACATTCGGCTACAGATACCAAAAAAAAAAAAAA
AAAAAATTTTCACCATTGAGGCACCACCTTCT
CGTCGCTGCGTCGCTCTGCTCGCGGCTAAAAA
ATTA**GAAACA**ACATTCGGCTACAGATACCAAA
ATTT

“Fuzzy” matching is much more computationally expensive.

Problems

A sequencer can produce over
1 billion reads per run

About 50% of the genome is made
of repetitive sequences

Sequencers make lots of mistakes

Every human has ~3 million SNPs
compared to the reference genome

Hash-based mapping:

Step1: hash/index the genome

Toy
genome
(16 bp)

CATGGTCATTGGTTCC

Hash-based mapping:

Step1: hash/index the genome

CATGGTCATTGGTTCC

k = 3

Kmer/Hash

Genome Positions

CAT

1

Hash-based mapping:

Step1: hash/index the genome

CATGGTCATTGGTTCC

k = 3

Kmer/Hash

Genome Positions

CAT
ATG

1
2

Hash-based mapping:

Step1: hash/index the genome

CATGGTCATTGGTTCC

k = 3

Kmer/Hash

Genome Positions

CAT
ATG
TGG

1
2
3

Hash-based mapping:

Step1: hash/index the genome

CATGGTCATTGGTTCC

k = 3

Kmer/Hash

Genome Positions

CAT
ATG
TGG
GGT

1
2
3
4

Hash-based mapping:

Step1: hash/index the genome

CATG**GTC**ATTGGTTCC

k = 3

Kmer/Hash

Genome Positions

CAT
ATG
TGG
GGT
GTC

1
2
3
4
5

Hash-based mapping:

Step1: hash/index the genome

CATGGTCAATTGGTTCC

k = 3

Kmer/Hash

Genome Positions

CAT
ATG
TGG
GGT
GTC
TCA

1
2
3
4
5
6

Hash-based mapping:

Step1: hash/index the genome

CATGGT**CAT**TGGTTCC

k = 3

Kmer/Hash

Genome Positions

CAT
ATG
TGG
GGT
GTC
TCA

1, 7
2
3
4
5
6

Hash-based mapping:

Step1: hash/index the genome

CATGGTCATTGGTTCC

k = 3

Kmer/Hash

Genome Positions

CAT

1, 7

ATG

2

TGG

3, 10

GGT

4, 11

GTC

5

TCA

6

ATT

8

TTG

9

GTT

12

TTC

13

TCC

14

Complete hash/kmer index of our toy genome (forward strand only)

Hash-based mapping:

Step2: use the index to map (i.e., find alignment locations) reads

Toy
genome

CATGGTCATTGGTTCC

Kmer/Hash

Genome Positions

CAT	1, 7
ATG	2
TGG	3, 10
GGT	4, 11
GTC	5
TCA	6
ATT	8
TTG	9
GTT	12
TTC	13
TCC	14

Read

TGGTCA



kmer index is used to quickly find candidate alignment locations in genome.

Hash-based mapping:

Step2: use the index to map (i.e., find alignment locations) reads

Toy
genome

CATGGTCATTGGTTCC

Kmer/Hash

Genome Positions

CAT

1, 7

ATG

2

TGG

3, 10

GGT

4, 11

GTC

5

TCA

6

ATT

8

TTG

9

GTT

12

TTC

13

TCC

14

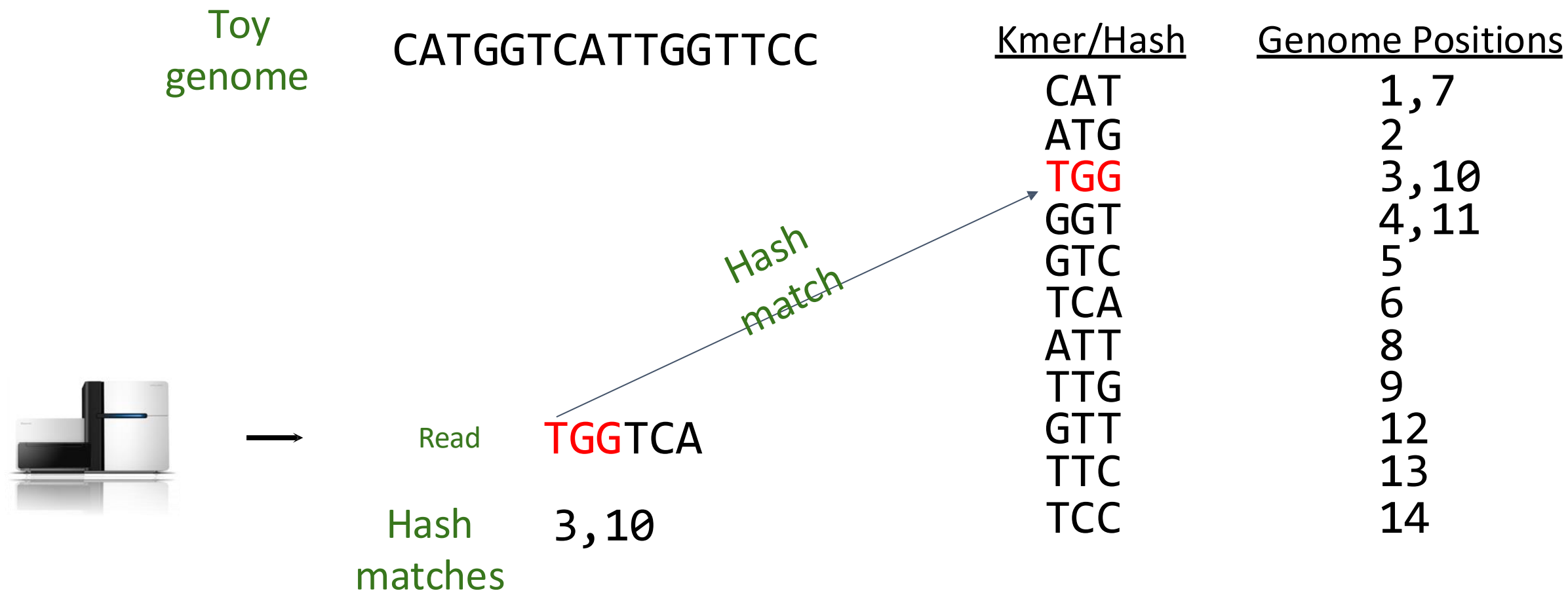


Read

TGGTCA

Hash-based mapping:

Step2: use the index to map (i.e., find alignment locations) reads



Hash-based mapping:

Step2: use the index to map (i.e., find alignment locations) reads



Toy
genome

CATGGTCATTGGTTCC

Kmer/Hash

Genome Positions

CAT

1, 7

ATG

2

TGG

3, 10

GGT

4, 11

GTC

5

TCA

6

ATT

8

TTG

9

GTT

12

TTC

13

TCC

14

Read

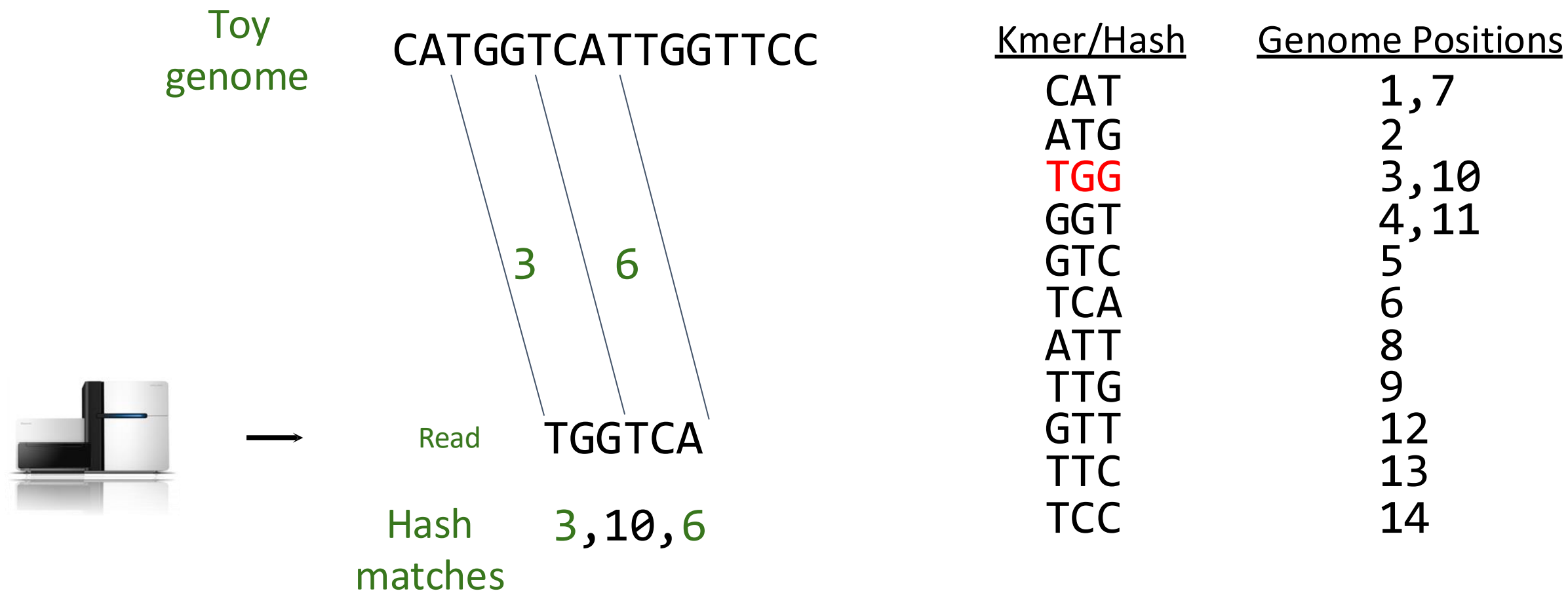
TGG**TCA**

Hash
matches

3, 10, 6

Hash-based mapping:

Step2: use the index to map (i.e., find alignment locations) reads



Hash-based mapping:

Step2: use the index to map (i.e., find alignment locations) reads

Toy
genome

CATGGTCATTGGTTCC

Kmer/Hash

Genome Positions

CAT

1, 7

ATG

2

TGG

3, 10

GGT

4, 11

GTC

5

TCA

6

ATT

8

TTG

9

GTT

12

TTC

13

TCC

14



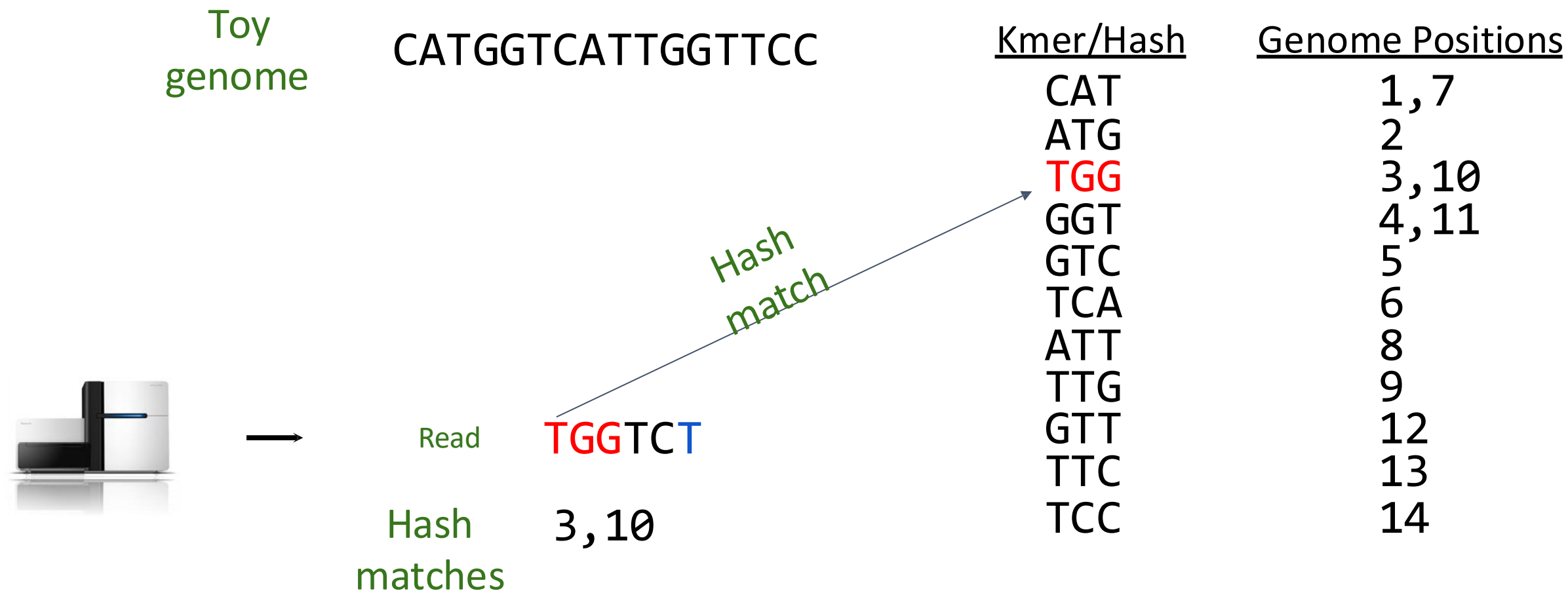
Read

TGGTCT

Even if we have SNPs, we can use the rest of the read to find plausible locations, then do more expensive exact comparisons

Hash-based mapping:

Step2: use the index to map (i.e., find alignment locations) reads



Alignment inputs

Reference genome (grch38.fa)

```
>chr1
TACCTCCAGGGGGGCATCCTCCCCCAATTC
GAAACACAATCGTAGCCCTGGCACTACCTA
TGTGTGTCAATTGGGAGAGAGAGATTAC
GAAAAAAAGTCTGGACTCACTAGGATACA
CACATTCGGCTACAGATACCAAAAAAAAAA
```



bwa index



Reference index

```
grch38.fa.amb
grch38.fa.ann
grch38.fa.bwt
grch38.fa.pac
grch38.fa.sa
```

Step 1: Building the index

This takes a long time,
(and a lot of memory)
but you only do it once!

Alignment algorithms

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jmb

Journal of Molecular Biology

Volume 48, Issue 3, 28 March 1970, Pages 443-453



A general method applicable to the search for similarities in the amino acid sequence of two proteins ☆

Saul B. Needleman, Christian D. Wunsch

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[https://doi.org/10.1016/0022-2836\(70\)90057-4](https://doi.org/10.1016/0022-2836(70)90057-4)

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Identification of Common Molecular Subsequences

T. F. SMITH AND M. S. WATERMAN

J. Mol. Biol. (1981), **147**, 195-197

Identification of Common Molecular Subsequences

The identification of maximally homologous subsequences among sets of long sequences is an important problem in molecular sequence analysis. The problem is straightforward only if one restricts consideration to contiguous subsequences (segments) containing no internal deletions or insertions. The more general problem has its solution in an extension of sequence metrics (Sellers 1974; Waterman *et al.*, 1976) developed to measure the minimum number of "events" required to convert one sequence into another.

These developments in the modern sequence analysis began with the heuristic homology algorithm of Needleman & Wunsch (1970) which first introduced an iterative matrix method of calculation. Numerous other heuristic algorithms have been suggested including those of Fitch (1966) and Dayhoff (1969). More mathematically rigorous algorithms were suggested by Sankoff (1972), Reichert *et al.* (1973) and Beyer *et al.* (1979), but these were generally not biologically satisfying or interpretable. Success came with Sellers (1974) development of a true metric measure of the distance between sequences. This metric was later generalized by Waterman *et al.* (1976) to include deletions/insertions of arbitrary length. This metric represents the minimum number of "mutational events" required to convert one sequence into another. It is of interest to note that Smith *et al.* (1980) have recently shown that under some conditions the generalized Sellers metric is equivalent to the original homology algorithm of Needleman & Wunsch (1970).

In this letter we extend the above ideas to find a pair of segments, one from each of two long sequences, such that there is no other pair of segments with greater similarity (homology). The similarity measure used here allows for arbitrary length deletions and insertions.

Sequence alignment software

	<u>Aligner</u>	<u>Applications</u>	<u>Availability</u>
→	BWA mem	DNA-seq	open-source
	Bowtie2	DNA-seq	open-source
	Novoalign	DNA-seq	free for academic use
	HISAT	RNA-seq	open-source
→	STAR	RNA-seq	open-source
→	minimap2	long-read DNA/RNA	open-source

BWA-MEM

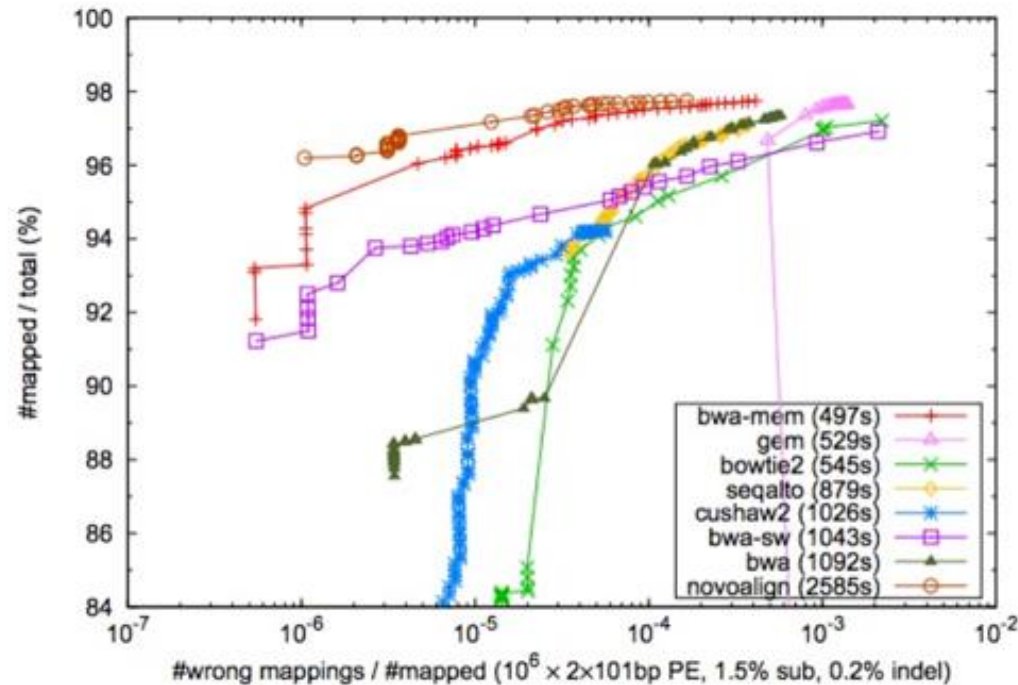


Fig. 1. Percent mapped reads as a function of the false alignment rate under different mapping quality cutoff. Alignments with mapping quality 3 or lower are excluded. An alignment is *wrong* if after correcting clipping, its start position is within 20bp from the simulated position. 10^6 pairs of 101bp reads are simulated from the human reference genome using wgsim (<http://bit.ly/wgsim2>) with 1.5% substitution errors and 0.2% indel variants. The insert size follows a normal distribution $N(500, 50^2)$. The reads are aligned back to the genome either as single end (SE; top panel) or as paired end (PE; bottom panel). GEM is configured to allow up to 5 gaps and to output suboptimal alignments (option '-e5 -m5 -s1' for SE and '-e5 -m5 -s1 -pb' for PE). GEM does not compute mapping quality. Its mapping quality is estimated with a BWA-like algorithm with suboptimal alignments available. Other mappers are run with the default setting except for specifying the insert size distribution. The run time in seconds on a single CPU core is shown in the parentheses.

Aligning sequence reads, clone sequences and assembly contigs with BWA-MEM

Heng Li

Broad Institute of Harvard and MIT, 7 Cambridge Center, Cambridge, MA 02142, USA

<https://arxiv.org/pdf/1303.3997v2.pdf>

Alignment inputs

Reference genome (grch38.fa)

```
>chr1
TACCTCCAGGGGGCATCCTCCCCCAATTC
GAAACACAATCGTAGCCCTGGCACTACCTA
TGTGTGTCAATTCGGAGAGAGAGATTAC
GAAAAAAAGTCTGGACTCACTAGGATACA
CACATTCGGCTACAGATACCAAAAAAAAAA
```



bwa index



Reference index

```
grch38.fa.amb
grch38.fa.ann
grch38.fa.bwt
grch38.fa.pac
grch38.fa.sa
```



Unaligned Sample Data In FASTQ format

```
@seq1
ATTCGAAACACGAAA...
+
DDED88 (99967DA...
```



Alignment inputs

Reference genome (grch38.fa)

```
>chr1
TACCTCCAGGGGGCATCCTCCCCCAATTC
GAAACACAATCGTAGCCCTGGCACTACCTA
TGTGTGTCAATTCGGAGAGAGAGATTAC
GAAAAAAAGTCTGGACTCACTAGGATACA
CACATTCGGCTACAGATACCAAAAAAAAAA
```



bwa index



Reference index

```
grch38.fa.amb
grch38.fa.ann
grch38.fa.bwt
grch38.fa.pac
grch38.fa.sa
```

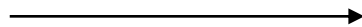


Unaligned Sample Data In FASTQ format

```
@seq1
ATTCGAAACACGAAA...
+
DDED88 (99967DA...
```



BWA
MEM



Step 2: Finding the matches

Alignment inputs

Reference genome
(grch38.fa)

```
>chr1
TACCTCCAGGGGGCATCCTCCCCCAATTC
GAAACACAATCGTAGCCCTGGCACTACCTA
TGTGTGTCAATT CGGAGAGAGAGATT CAC
GAAAAAAAAGTCTGGACTCAACTAGGATACA
CACATT CGGCTACAGATACCAAAAAAAAAA
```

bwa index

Reference index

grch38.fa.amb
grch38.fa.ann
grch38.fa.bwt
grch38.fa.pac
grch38.fa.sa



Unaligned
Sample Data
In FASTQ format

```
@seq1
ATTCGAAACACGAAA...
+
DDED88 (99967DA...
```

BWA MEM

Aligned Sample Data in SAM format

```
seq1  97  1  10001  0  4S15M1I54M2I50M25S
4  699063  0
CCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCC
TAACCCTAACCCCTAACCCCTAACCCCTAACCCCTAAACCCCTA
ACCCTAACCCCTAACCCCTAACCCCTAACCCCTAACCCAACCCT
AACCC
AAFFFKKKKKKKKKKKKKKKKKKKKKKKFKKKFKKKKKKKFKKAK
KKKFFKKKKKFKF<FF7FFFFK7FK,AA,FKFFKF',FKKK
7<KKK,,7FF7F,AAFFK,AKA,,FFF<,A7FKK,,7A
,,,7,,
```

SAM format: a text-based standard for representing sequence alignments

BIOINFORMATICS APPLICATIONS NOTE Vol. 25 no. 16 2009, pages 2078–2079
doi:10.1093/bioinformatics/btp352

Sequence analysis

The Sequence Alignment/Map format and SAMtools

Heng Li^{1,†}, Bob Handsaker^{2,†}, Alec Wysoker², Tim Fennell², Jue Ruan³, Nils Homer⁴, Gabor Marth⁵, Goncalo Abecasis⁶, Richard Durbin^{1,*} and 1000 Genome Project Data Processing Subgroup⁷

¹Wellcome Trust Sanger Institute, Wellcome Trust Genome Campus, Cambridge, CB10 1SA, UK, ²Broad Institute of MIT and Harvard, Cambridge, MA 02141, USA, ³Beijing Institute of Genomics, Chinese Academy of Science, Beijing 100029, China, ⁴Department of Computer Science, University of California Los Angeles, Los Angeles, CA 90095, ⁵Department of Biology, Boston College, Chestnut Hill, MA 02467, ⁶Center for Statistical Genetics, Department of Biostatistics, University of Michigan, Ann Arbor, MI 48109, USA and ⁷<http://1000genomes.org>

Received on April 28, 2009; revised on May 28, 2009; accepted on May 30, 2009

Advance Access publication June 8, 2009

Associate Editor: Alfonso Valencia

Table 1. Mandatory fields in the SAM format

No.	Name	Description
1	QNAME	Query NAME of the read or the read pair
2	FLAG	Bitwise FLAG (pairing, strand, mate strand, etc.)
3	RNAME	Reference sequence NAME
4	POS	1-Based leftmost POSition of clipped alignment
5	MAPQ	MAPping Quality (Phred-scaled)
6	CIGAR	Extended CIGAR string (operations: MIDNSHP)
7	MRNM	Mate Reference NaMe ('=' if same as RNAME)
8	MPOS	1-Based leftmost Mate POSition
9	ISIZE	Inferred Insert SIZE
10	SEQ	Query SEQUENCE on the same strand as the reference
11	QUAL	Query QUALity (ASCII-33=Phred base quality)

SAM format header

<http://samtools.sourceforge.net/samtools.shtml>

```
@HD VN:1.5 SO:coordinate
@SQ SN:1 LN:195471971
@SQ SN:10 LN:130694993
@SQ SN:11 LN:122082543
@SQ SN:12 LN:120129022
@SQ SN:13 LN:120421639
@SQ SN:14 LN:124902244
@SQ SN:15 LN:104043685
...
```

Contains metadata about what reference genome was used and what tools were run to create the file

```
@RG ID:1bc1076b8d1c41bd92a141830ea648a7 CN:WUGSC LB:M-RB_60975_DNMT3A-
R878H_BM_3A-ab_ChIPseq-lib1 PL:Illumina
PU:1bc1076b8d1c41bd92a141830ea648a7.unknown SM:M-RB_60975_DNMT3A-R878H_BM_3A-
ab_ChIPseq
@PG ID:1bc1076b8d1c41bd92a141830ea648a7 PN:bwa CL:/usr/local/bin/bwa mem -K
100000000 -t 8 -Y -R @RG VN:0.7.15-r1140
@PG ID:SAMBLASTER CL:samblaster -i stdin -o stdout --acceptDupMarks --
addMateTags VN:0.1.24
```

[illegible]

SAM format body

<http://samtools.sourceforge.net/samtools.shtml>

Col #	Name	Meaning	Example
1	QNAME	Read or Pair name	HWI-ST156_1:278:1:1058:4544:0
2	FLAG	Bitwise FLAG	alignment status
3	RNAME	Reference sequence name	chr1
4	POS	1-based alignment start coordinate	8,724,005
5	MAPQ	Mapping quality	60
6	CIGAR	Extended CIGAR string	local alignment info
7	MRNM	If paired, the mate's reference seq.	chr1
8	MPOS	If paired, the mate's alignment start	8,724,505
9	ISIZE	If paired, the insert size	562
10	SEQ	The sequence of the query/mate	ACAAATTCAG...
11	QUAL	The quality string for the query/mate	HHH\$^^%\$\$\$...
12	OPT	Optional Tags	XA:i:2, MD:Z:0T34G15

SAM format overview

<http://samtools.sourceforge.net/samtools.shtml>

Col #	Name	Meaning	Example
1	QNAME	Read or Pair name	HWI-ST156_1:278:1:1058:4544:0
2	FLAG	Bitwise FLAG	alignment status
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8	MPOS	If paired, the mate's alignment start	8,724,505
9	ISIZE	If paired, the insert size	562
10	SEQ	The sequence of the query/mate	ACAAATTCAG...
11	QUAL	The quality string for the query/mate	HHH\$^^%\$\$\$...
12	OPT	Optional Tags	XA:i:2, MD:Z:0T34G15

straight from the FASTQ file

*Unless you've done something like trimming or base-quality recalibration (BQSR)

SAM format overview

<http://samtools.sourceforge.net/samtools.shtml>

Col #	Name	Meaning	Example
1	QNAME	Read or Pair name	HWI-ST156_1:278:1:1058:4544:0
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11	QUAL	The quality string for the query/mate	HHH\$^^%\$\$\$...
12	OPT	Optional Tags	XA:i:2, MD:Z:0T34G15

Genome coordinates
chr/start position

SAM format overview

<http://samtools.sourceforge.net/samtools.shtml>

Col #	Name	Meaning	Example
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6	CIGAR	Extended CIGAR string	local alignment info
7	MRNM	If paired, the mate's reference seq.	chr1
8	MPOS	If paired, the mate's alignment start	8,724,505
9	ISIZE	If paired, the insert size	562
10	SEQ	The sequence of the query/mate	ACAAATTCAG...
11	QUAL	The quality string for the query/mate	HHH\$^^%\$\$\$...
12	OPT	Optional Tags	XA:i:2, MD:Z:0T34G15

How confident is the aligner in this mapping?

Uses the same Phred scale as base quality

$$\text{MAPQ} = -10 \cdot \log_{10}(P_{\text{map_loc_wrong}})$$

$(P_{\text{map_loc_wrong}})$	$\log_{10}(P_{\text{map_loc_wrong}})$	MAPQ
--------------------------------	---	------

1	0	0
0.1	-1	10
0.01	-2	20
0.001	-3	30
0.0001	-4	40

SAM format overview

<http://samtools.sourceforge.net/samtools.shtml>

Col #	Name	Meaning	Example
1	QNAME	Read or Pair name	HWI-ST156_1:278:1:1058:4544:0
2	FLAG	Bitwise FLAG	alignment status
3	RNAME	Reference sequence name	chr1
4	POS	1-based alignment start coordinate	8,724,005
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6	CIGAR	Extended CIGAR string	local alignment info
7	MRNM	If paired, the mate's reference seq.	chr1
8	MPOS	If paired, the mate's alignment start	8,724,505
9	ISIZE	If paired, the insert size	562
10	SEQ	The sequence of the query/mate	ACAAATTCAG...
11	QUAL	The quality string for the query/mate	HHH\$^^%\$\$\$...
12	OPT	Optional Tags	XA:i:2, MD:Z:0T34G15

basepair alignment details

The CIGAR string: encode the details of the alignment

Operation	Meaning
M	Match*
D	Deletion w.r.t. reference
I	Insertion w.r.t. reference
N	Split or spliced alignment
S	Soft-clipping
H	Hard-clipping
P	Padding

Reference:
Experimental:

ACCTGTC--TACCTTACG
ACCT-TCCATACTTTATC



4M 1D2M 2I 7M 2S

CIGAR string:

4M1D2M2I7M2S



LENGTH/OPERATION

SAM format overview

<http://samtools.sourceforge.net/samtools.shtml>

Col #	Name	Meaning	Example
1	QNAME	Read or Pair name	HWI-ST156_1:278:1:1058:4544:0
2	FLAG	Bitwise FLAG	alignment status
3	RNAME	Reference sequence name	chr1
4	POS	1-based alignment start coordinate	8,724,005
5	MAPQ	Mapping quality	60
6	CIGAR	Extended CIGAR string	local alignment info
7	MRNM	If paired, the mate's reference seq.	chr1
8	MPOS	If paired, the mate's alignment start	8,724,505
9	ISIZE	If paired, the insert size	562
10	SEQ	The sequence of the query/mate	ACAAATTCAG...
11	QUAL	The quality string for the query/mate	HHH\$^^%\$\$\$...
12	OPT	Optional Tags	XA:i:2, MD:Z:0T34G15

Alignment
characteristics
of this read and
its mate

FLAG

97
1123
99
99
99
81
99
99
163
99

The FLAG score

base2	base10	base16	Meaning	Applies to:
000000000001	1	0x0001	The read originated from a paired sequencing molecule	Both
000000000010	2	0x0002	The read is mapped in a proper pair	Pairs only
000000000100	4	0x0004	The query sequence itself is unmapped	Both
000000010000	8	0x0008	The query's mate is unmapped	Pairs only
000000100000	16	0x0010	Strand of the query (0 for forward; 1 for reverse strand)	Both
000001000000	32	0x0020	Strand of the query's mate	Pairs only
000010000000	64	0x0040	The query is the first read in the pair	Pairs only
000100000000	128	0x0080	The read is the second read in the pair	Pairs only
001000000000	256	0x0100	The alignment is not primary	Both
010000000000	512	0x0200	The read fails platform/vendor quality checks	Both
100000000000	1024	0x0400	The read is either a PCR duplicate or an optical duplicate	Both

ST-E00223:32:H5J57CCXX:4:1220:14651:8868

99

1

10086

base2	base10	base16	Meaning	Applies to:
0000000001	1	0x0001	The read originated from a paired sequencing molecule	Both
0000000010	2	0x0002	The read is mapped in a proper pair	Pairs only
0000000100	4	0x0004	The query sequence itself is unmapped	Both
0000001000	8	0x0008	The query's mate is unmapped	Pairs only
0000010000	16	0x0010	Strand of the query (0 for forward; 1 for reverse strand)	Both
0000100000	32	0x0020	Strand of the query's mate	Pairs only
0001000000	64	0x0040	The query is the first read in the pair	Pairs only
0010000000	128	0x0080	The read is the second read in the pair	Pairs only
0100000000	256	0x0100	The alignment is not primary	Both
1000000000	512	0x0200	The read fails platform/vendor quality checks	Both
1000000000	1024	0x0400	The read is either a PCR duplicate or an optical duplicate	Both

00001100011

$$2^6 + 2^5 + 2^1 + 2^0 = 64 + 32 + 2 + 1 = 99$$

ST-E0022

36

base2
00000000001
00000000010
00000000100
00000001000
00000010000
00000100000
00001000000
00010000000
00100000000
01000000000
10000000000

SAM Flag: Explain

Switch to mate Toggle first in pair / second in pair

Find SAM flag by property:

To find out what the SAM flag value would be for a given combination of properties, tick the boxes for those that you'd like to include. The flag value will be shown in the SAM Flag field above.

- ☒ read paired
- ☒ read mapped in proper pair
- ☐ read unmapped
- ☐ mate unmapped
- ☐ read reverse strand
- ☒ mate reverse strand
- ☒ first in pair
- ☐ second in pair
- ☐ not primary alignment
- ☐ read fails platform/vendor quality checks
- ☐ read is PCR or optical duplicate
- ☐ supplementary alignment

Summary:

- read paired (0x1)
- read mapped in proper pair (0x2)
- mate reverse strand (0x20)
- first in pair (0x40)

Applies to:
only
only
only
only
only

$$2^6 + 2^5 + 2^1 + 2^0 = 64 + 32 + 2 + 1 = 99$$

SAM format overview

<http://samtools.sourceforge.net/samtools.shtml>

Col #	Name	Meaning	Example
1	QNAME	Read or Pair name	HWI-ST156_1:278:1:1058:4544:0
2	FLAG	Bitwise FLAG	alignment status
3	RNAME	Reference sequence name	chr1
4	POS	1-based alignment start coordinate	8,724,005
5	MAPQ	Mapping quality	60
6	CIGAR	Extended CIGAR string	local alignment info
7	MRNM	If paired, the mate's reference seq.	chr1
8	MPOS	If paired, the mate's alignment start	8,724,505
9	ISIZE	If paired, the insert size	562
10	SEQ	The sequence of the query/mate	ACAAATTCAG...
11	QUAL	The quality string for the query/mate	HHH\$^^%\$\$\$...
12	OPT	Optional Tags	XA:i:2, MD:Z:0T34G15

Where the mate pair
read aligns

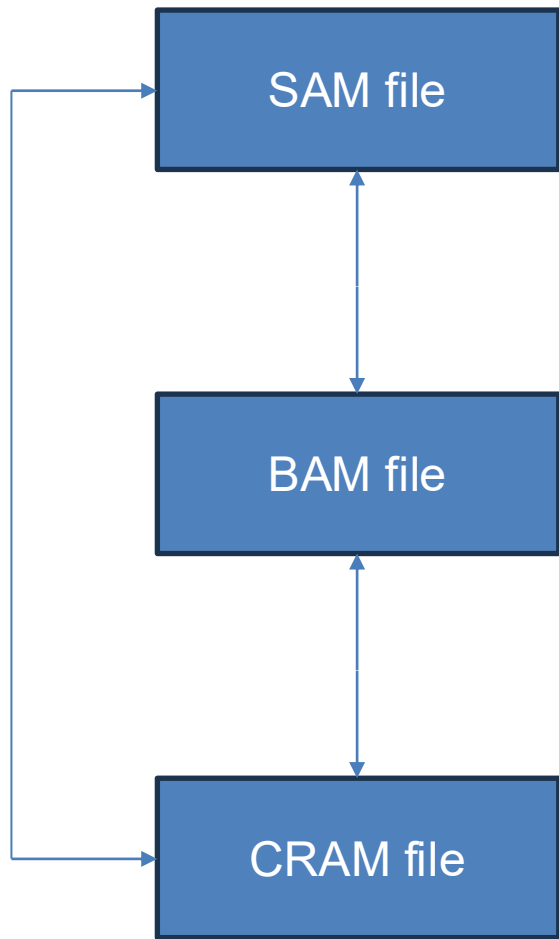
How close is the mate to the
first read

SAM format overview

<http://samtools.sourceforge.net/samtools.shtml>

Col #	Name	Meaning	Example
1	QNAME	Read or Pair name	HWI-ST156_1:278:1:1058:4544:0
2	FLAG	Bitwise FLAG	alignment status
3	RNAME	Reference sequence name	chr1
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5	MAPQ	Mapping quality	60
6	CIGAR	Extended CIGAR string	local alignment info
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8	MPOS	If paired, the mate's alignment start	8,724,505
9	ISIZE	If paired, the insert size	562
10	SEQ	The sequence of the query/mate	ACAAATTCAG...
11	QUAL	The quality string for the query/mate	HHH\$^^%\$\$\$...
12	OPT	Optional Tags	XA:i:2, MD:Z:0T34G15

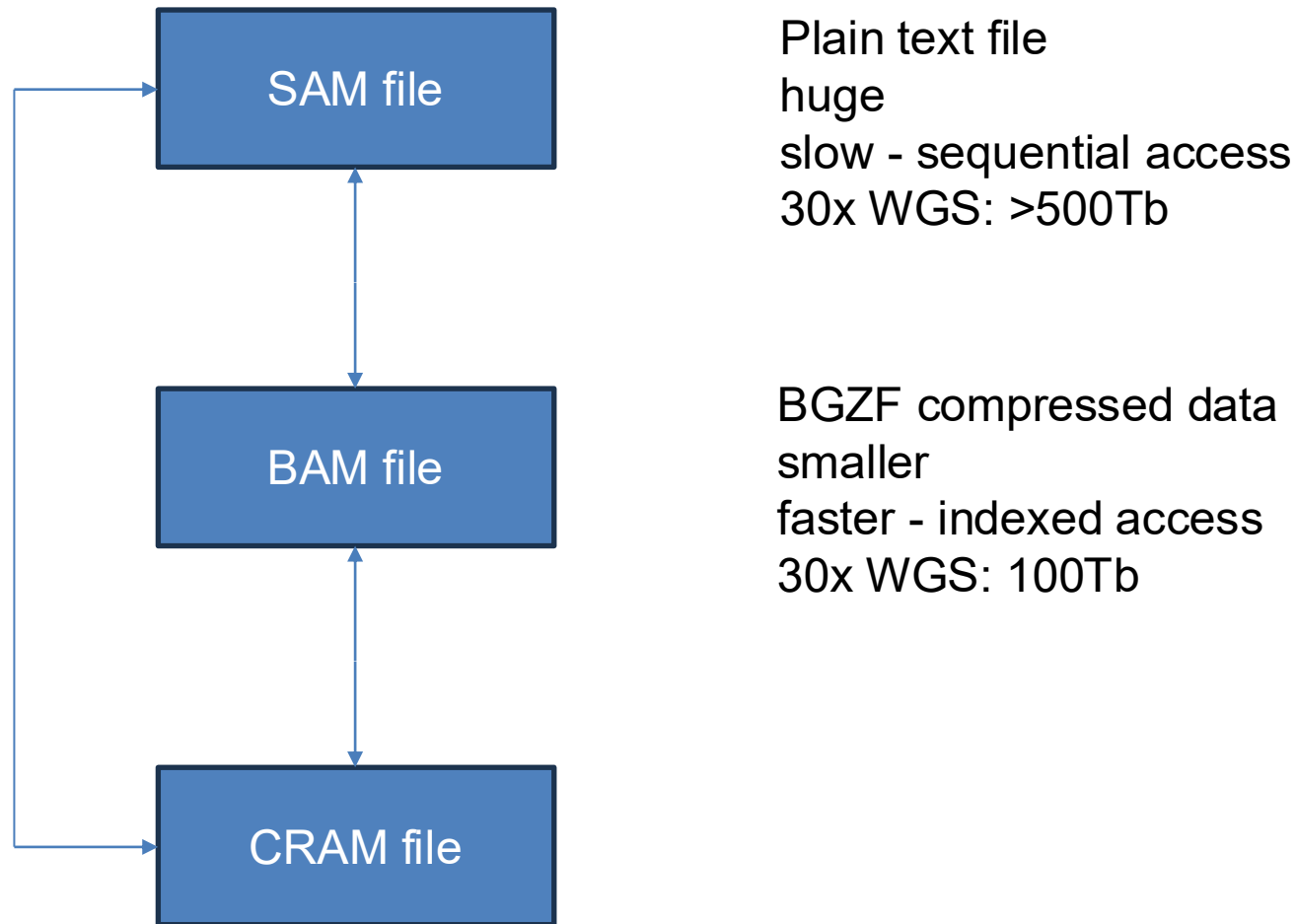
SAM, BAM, and CRAM files



Plain text file
huge
slow - sequential access
30x WGS: >500Tb

We almost never store data in
SAM files - it's huge!

SAM, BAM, and CRAM files



Plain text file
huge
slow - sequential access
30x WGS: >500Tb

BGZF compressed data
smaller
faster - indexed access
30x WGS: 100Tb

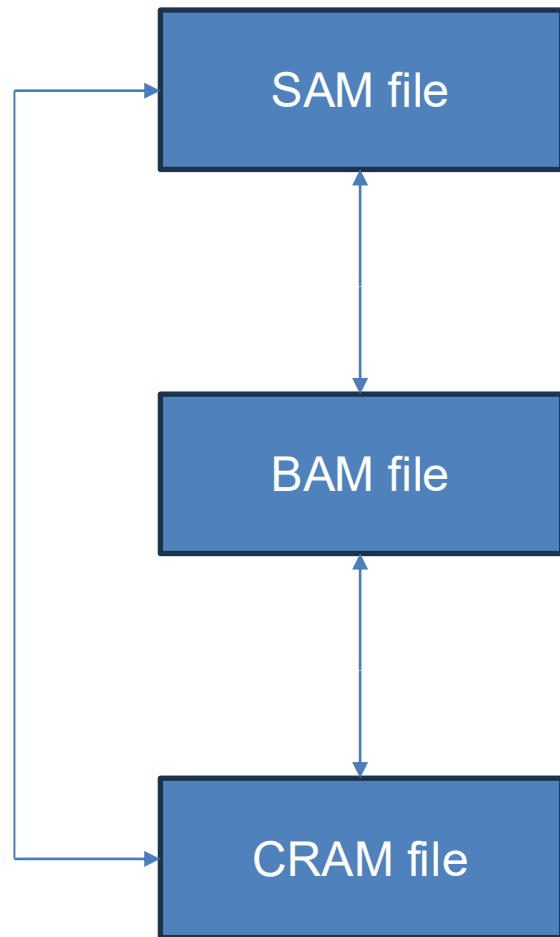
BAMs usually have an **index file**:
like a table of contents that allows
skipping to the right section

(only works if your bam is *sorted*)

The index lives next to the bam in
a separate file:

`myreads.bam`
`myreads.bam.bai`

SAM, BAM, and CRAM files



Plain text file
huge
slow - sequential access
30x WGS: >500Tb

BGZF compressed data
smaller
faster - indexed access
30x WGS: 100Tb

Reference-based compression
smallest
fast – some overhead for decompression
requires a reference for decoding
30x WGS: 50Tb

CRAMs get smaller by storing the *difference* from the reference genome

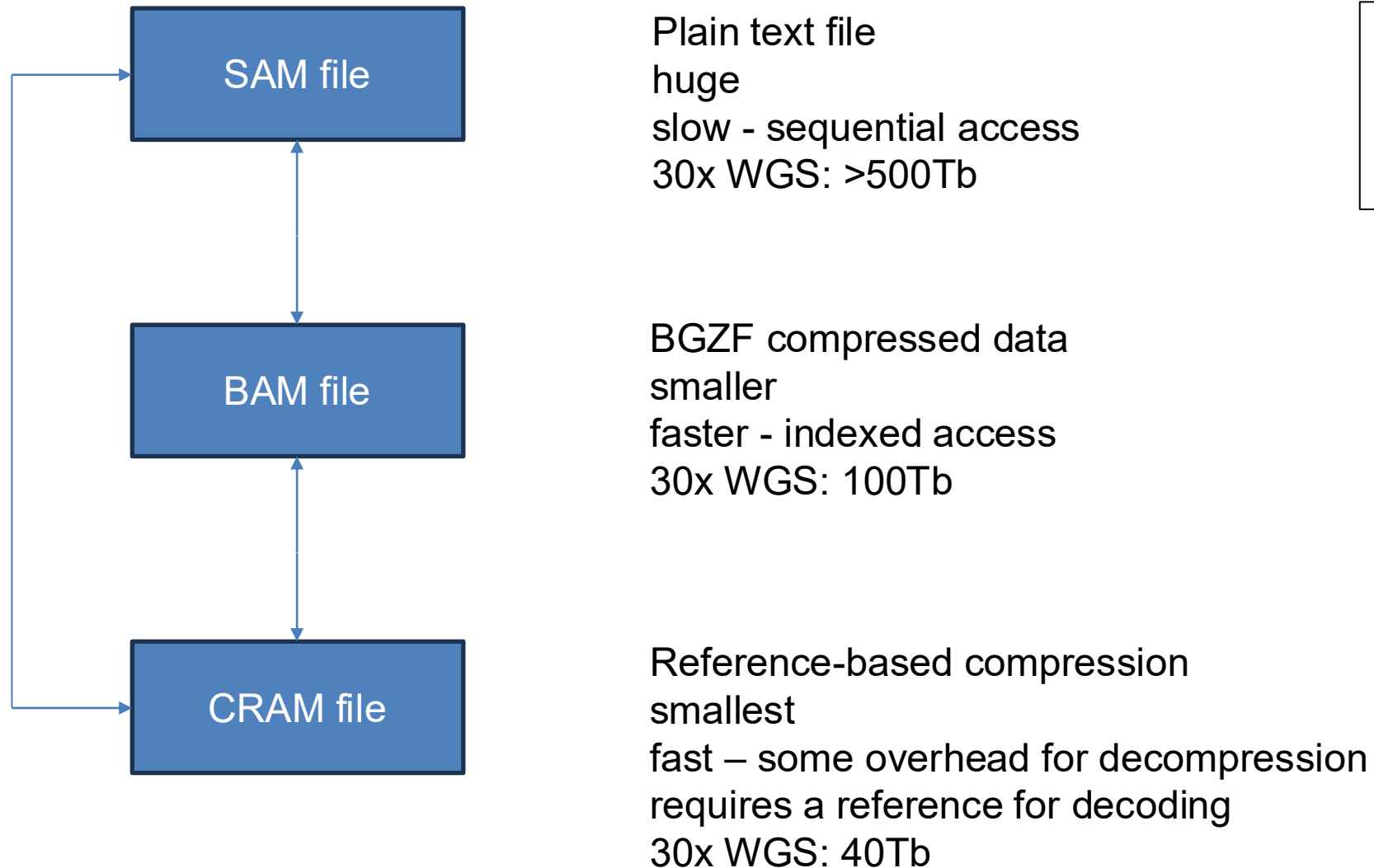
CRAMs require access to the reference genome to use the data

Have to be able to undo the reference compression!

Also has an index that lives next to it in a separate file:

`myreads.cram`
`myreads.cram.crai`

SAM, BAM, and CRAM files



convert between these
formats using **samtools**

Mounting cluster storage to your laptop

IGV intro